

Text Preprocessing

Week 3 – Programming

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❖ Overview

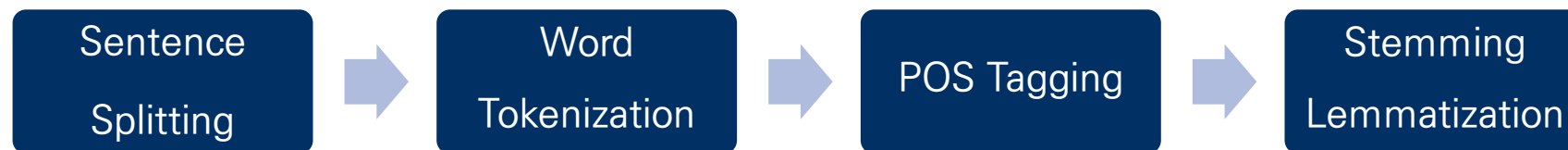
- Teaching Assistant: Minjoo Son
- Email: minjoo77@hanyang.ac.kr
 - Please feel free to ask me anytime if you have any questions!

❖ Week 4 Objective

- What we've learned in the past lectures?
 - Overview of Unstructured Data Analysis
 - Text Analytics Process
 - Text Preprocessing
- What will we do in today's lectures?
 - Practicing basic text preprocessing methods in code
 - Text Preprocessing Process
 - Visualization
 - Self-Guided Practice

❖ Text Preprocessing

- Purpose: Prepares text according to the problem's purpose
- Analogy: Just like how a dish turns out poorly if the ingredients are not properly prepared, natural language processing techniques learned later may not work correctly if the text is not properly preprocessed.
- Process:

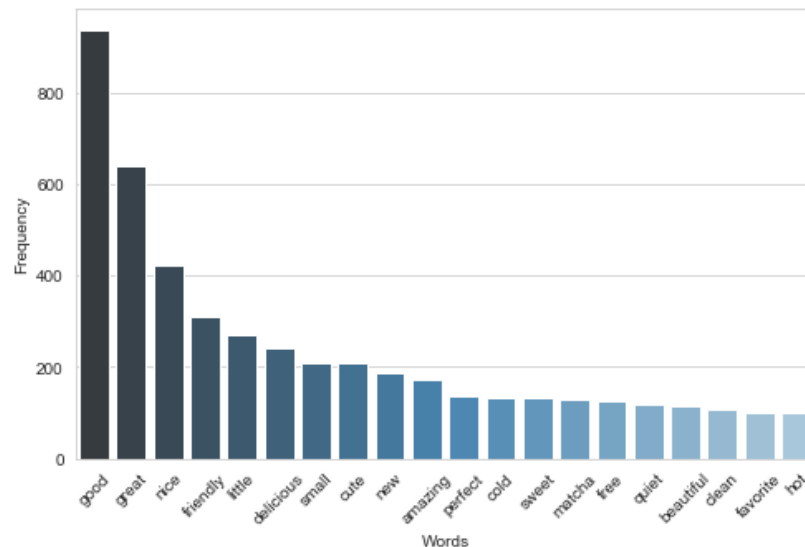


❖ Programming

- Programming in Colab
- Link
 - https://drive.google.com/file/d/18hyIYY1Y2qpN-IgQKHC8Sx0IgTag2Wi6/view?usp=drive_link
(‘1. Text Preprocessing Process.ipynb’)

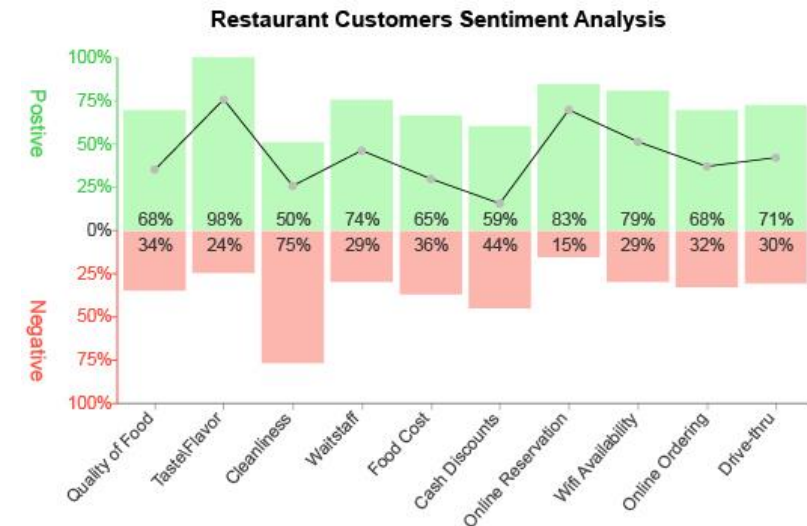
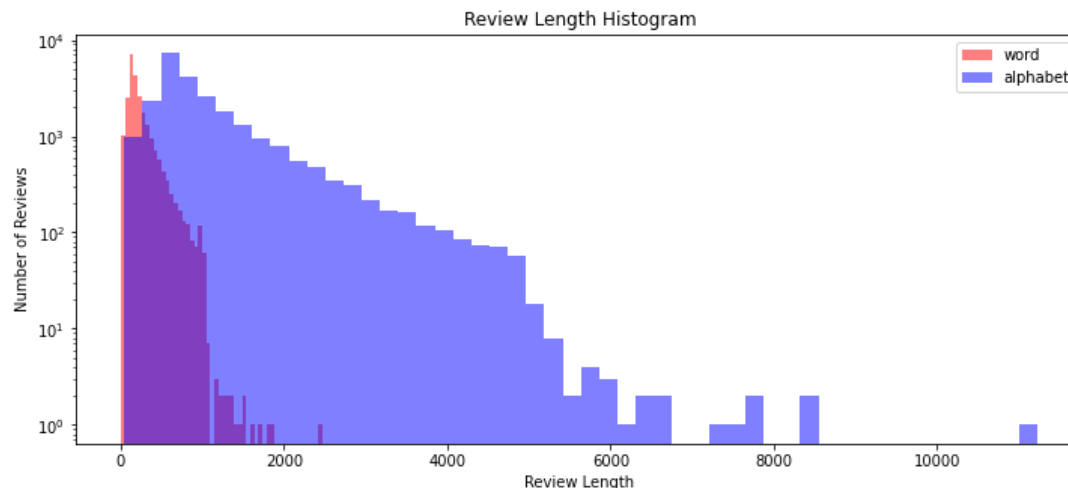
❖ Word Frequency Graph & Word Cloud

- Word Frequency Graph
 - A tool that visually identifies key words and patterns, helping to remove unnecessary words or select words for learning.
- Word Cloud
 - A visual representation of word frequency, allowing for easy identification of key concepts in the text.



❖ Word Frequency Graph & Word Cloud

- EDA (Exploratory Data Analysis)
 - The process of understanding data and identifying patterns and characteristics.
 - In Natural Language Processing, it is the initial step to understand text data.
 - It involves identifying the data structure, analyzing word frequencies and patterns, and visualizing them with a Word Cloud.
 - Sentiment analysis is used to detect positive or negative trends, while data distribution analysis captures overall characteristics.



❖ Programming

- Programming in Colab.
- Link
 - https://drive.google.com/file/d/1OoSaRvJRgJbMTIQ2bS1LILDAnPTofM9Z/view?usp=drive_link
[File](#)
(‘2. Visualization.ipynb’)

❖ Programming

- Text Preprocessing & EDA

- Task

- Preprocess the Twitter Sentiment Analysis dataset, then create visualizations like a Word Graph or Word Cloud to represent each sentiment (Positive, Negative, Neutral, exclude Irrelevant).

- Dataset

- This is an entity-level sentiment analysis dataset of twitter.
 - Entities: Borderlands, Microsoft, TomClancysRainbowSix, MaddenNFL, LeagueOfLegends, CallOfDuty
 - There are three classes in this dataset: Positive, Negative and Neutral. We regard messages that are not relevant to the entity (i.e. Irrelevant) as Neutral.
 - Data (Uploaded on Google Drive): Unstructured Data Analysis › [Week 3] Text Preprocessing › data › twcs.csv

- Process

- Load the data
 - Text preprocessing
 - Text Visualization

❖ Programming

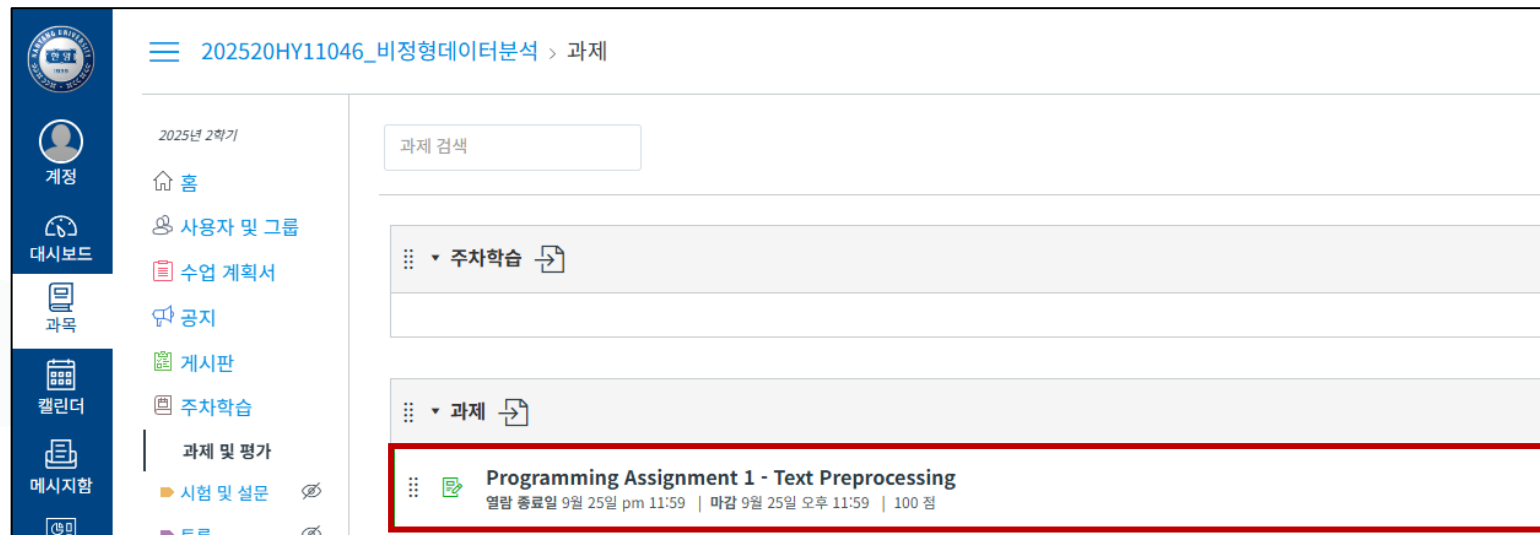
- Text Preprocessing & EDA

- Deliverables

- Jupyter Notebook file: Assignment_1_YourName_YourStudentID.ipynb ('Assignment_1_Minjoo Son_2024229436.ipynb')
 - Submit to: "Programming Assignment 1 – Text Preprocessing" under Assignments

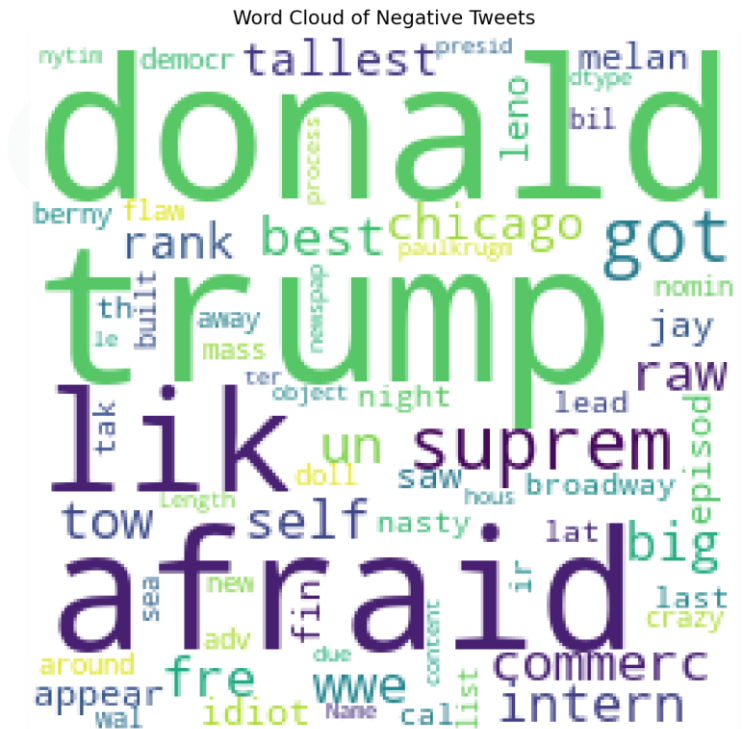
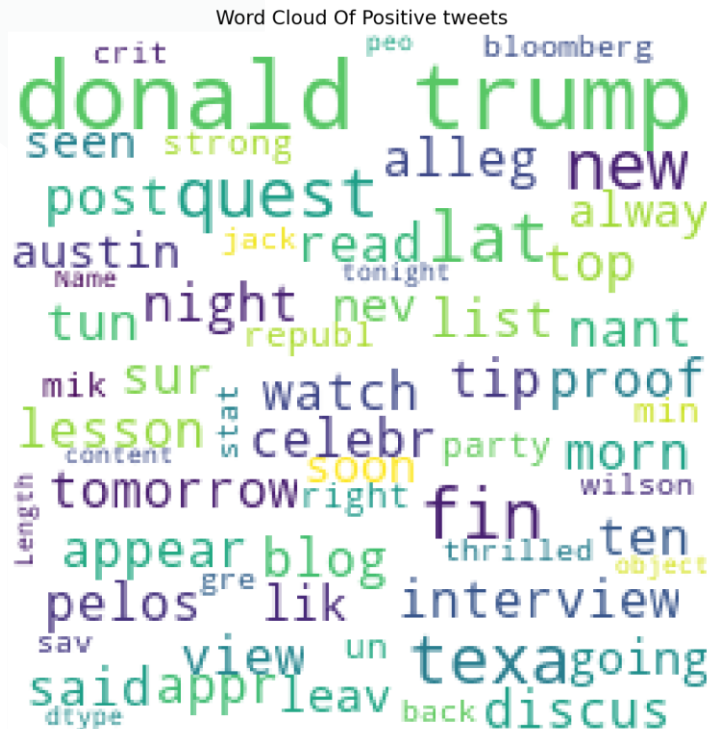
- Deadline

- September 25, 2025, 11:59 PM



❖ Programming

- Text Preprocessing & EDA
 - Example Results



❖ Programming

- There is a reference code on Colab, which you can use as a guide to complete the task.
- Link
 - https://drive.google.com/file/d/1DuTS9tMb4vKgwoolioed7YMS8ecBpXO6/view?usp=drive_link
('optional) text preprocessing sample.ipynb')

❖ Request for Feedback to Improve Future Classes

- The following is a survey for today's Programming class.
- Please provide feedback on today's class and share your preferred format for future sessions.
- The survey is anonymous, so feel free to express your thoughts openly.
- Your feedback will be used to improve the Programming class in future sessions.

Q & A

Thank you for your attention. Any questions are welcome!

Minjoo Son

